

UNIVERSITY OF CENTRAL MISSOURI

PROJECT REPORT

ON

**PREDICTING STOCK MARKET TRENDS
USING NLP & DEEP LEARNING**

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ABSTRACT

Stock market prediction is a challenging task influenced by both historical price trends and external factors such as financial news and public sentiment. Traditional models rely mainly on numerical data and often fail to capture the impact of such unstructured information. This project presents a sentiment-aware stock forecasting system that integrates Natural Language Processing (NLP) with machine learning techniques.

Sentiment is extracted from financial news using approaches like VADER and FinBERT, and combined with stock price data to build predictive models including LSTM and hybrid architectures. The hybrid approach enables the system to capture both time-series patterns and market sentiment.

Experimental results show that incorporating sentiment improves prediction accuracy and overall model performance compared to price-only models. This work highlights the importance of combining structured and unstructured data for more reliable and context-aware stock market forecasting.

Keywords: Stock Market Prediction, Natural Language Processing (NLP), Sentiment Analysis, LSTM, Hybrid Model, Financial News Analysis, Machine Learning, Multimodal Data Fusion, Time-Series Forecasting, FinBERT

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1.INTRODUCTION

Financial markets respond not just to numbers, but to stories. Earnings reports, geopolitical developments, regulatory announcements, and shifts in public confidence all leave traces in the news long before they register in a stock's price chart. Yet most traditional forecasting models treat markets as if they exist in an information vacuum, relying exclusively on historical price data to predict future direction. This project challenges that assumption.

The Sentiment-Aware Stock Forecasting system is built on a straightforward premise: if sentiment moves before prices do, then capturing sentiment from financial news should improve the accuracy and timeliness of market predictions. By combining structured time-series data specifically daily OHLCV (Open, High, Low, Close, Volume) records with unstructured text signals extracted from financial headlines, the system aims to close the gap between what the market knows and what the model sees.

To extract sentiment, two complementary tools are employed. VADER provides fast, rule-based polarity scoring suitable for high-volume processing. FinBERT, a BERT-based transformer pre-trained on financial corpora, delivers contextually nuanced sentiment classification that captures the domain-specific language of financial reporting. These sentiment signals are then fused with price data through a suite of machine learning architectures a Baseline LSTM, a Hybrid Fusion model, an NLP Tree Ensemble, and a Multi-Model Ensemble allowing the project to systematically evaluate how different integration strategies affect predictive performance.

1.1 Project Motivation

Financial markets react instantly to information, yet most traditional forecasting models only look at past prices. This creates a structural delay: news breaks first, prices adjust later, and models relying solely on technical indicators often miss that early window. By bringing NLP-based sentiment analysis into the pipeline, the project attempts to bridge this gap—measuring the prevailing market mood directly from headlines so the model can respond to important events before they fully show up in price data.

1.2 Importance and Relevance

In a trading environment where information travels faster than ever, understanding sentiment is becoming as operationally important as understanding price action. This work matters for three interconnected reasons:

- **News sentiment often leads to prices:** Major market moves are frequently preceded by observable shifts in news tone, making sentiment a valuable early signal of volatility that arrives before technical indicators can register it.
- **Combining structured and unstructured data improves model completeness:** Price charts capture what happened; news captures why it happened. Models that integrate both have a richer, more causally grounded view of market behavior.
- **Sentiment analysis strengthens risk management:** Sudden spikes in negative sentiment can flag rare, high-impact events before they fully materialize in prices, offering automated systems a critical early-warning mechanism.

1.3 Objectives and Research Questions

The project focuses on three main research questions:

- **Feature Engineering:** Can raw financial news be transformed into meaningful numerical sentiment features that carry predictive signal beyond what price data alone provides?
- **Model Architecture:** Which integration approach LSTM, tree-based models, or hybrid fusion—handles multimodal financial data most effectively in terms of directional accuracy?
- **Performance Impact:** Does adding sentiment improve key metrics Accuracy, Precision, and AUC-ROC compared to a price-only baseline, and by how much?

2.PROBLEM STATEMENT

Traditional stock-forecasting models mainly rely on historical price data and often ignore external factors like news, public sentiment, and global events. As a result, they struggle to predict sudden market movements driven by real-world developments.

2.1 Limitations of Existing Solutions

Conventional technical indicators and price-only LSTM models treat the market as if it exists in isolation, ignoring external triggers such as political events, regulatory decisions, and unexpected corporate announcements that can cause sudden price swings beyond what historical data can predict. These models are also reactive in nature, identifying trends only after price movements have already occurred, by which time the market has often absorbed the underlying news.

During intense news cycles or rare Black Swan events, price-only models frequently generate unstable or misleading predictions because they cannot interpret market sentiment such as fear, optimism, or uncertainty. As a result, they struggle to adapt to rapid sentiment-driven changes when accurate forecasting is most important.

2.2 The Integration Challenge

Price data is neatly structured and recorded at fixed daily intervals, while news arrives unpredictably throughout the trading day. Aligning these two data streams requires careful temporal engineering to ensure that each day's sentiment signal is correctly associated with the corresponding price window without introducing look-ahead bias. Many deep learning models including the Baseline LSTM used in this project tend to overfit the long-term bullish trend present in most historical equity data. This produces a degenerate 'predict UP every time' behavior: recall reaches 100%, but precision and AUC-ROC collapse, making the model useless as a practical forecasting tool.

2.3 Project Objective

This project aims to overcome these challenges by building a multimodal fusion model that blends NLP-derived sentiment with historical price data turning subjective market mood into a measurable feature that improves prediction accuracy, reduces false positives, and provides earlier warnings of potential reversals or high-impact events.

3.TOOLS & TECHNOLOGIES

The system is built on a modern, high-performance technology stack designed to handle the computational demands of deep learning and NLP.

Category	Tool / Library	Purpose
Deep Learning	TensorFlow	LSTM and Hybrid Fusion model training (GPU-accelerated)
Machine Learning	Scikit-learn	Preprocessing, NLP Tree Ensemble (Random Forest)
Gradient Boosting	XGBoost	NLP Tree Ensemble classifier (Model C)
NLP / Transformers	Hugging Face Transformers	FinBERT sentiment classification
NLP Utilities	NLTK & spaCy	Tokenization, lemmatization, Named Entity Recognition
Sentiment (Rule)	VADER	Fast rule-based polarity scoring
Financial Data	yfinance	Historical OHLCV price data retrieval
Data Handling	Pandas & NumPy	Time-series alignment, feature engineering
Visualisation	Matplotlib / Seaborn	Result charts, confusion matrices, SHAP plots
Deployment	Streamlit	Interactive prediction dashboard (app.py)
Development	VS Code	Scripting, EDA, rapid prototyping

4. METHODOLOGY

The methodology follows a clear, step-by-step workflow that transforms raw price data and unstructured news text into a unified multimodal forecasting system.

4.1 Data Collection & Sources

- **Market Data:** Historical OHLCV data gathered automatically using the yfinance library, creating a clean and consistent time-series foundation.
- **Textual Data:** Financial news headlines and short summaries collected from major financial outlets through RSS feeds and news APIs.

4.2 Data Preprocessing & Feature Engineering

Price data was checked for missing values and adjusted for corporate events such as stock splits and dividends. All numerical features were scaled using Min-Max normalization to the interval [0, 1], ensuring stable neural network training. Headlines were cleaned to remove HTML tags, punctuation, and formatting noise. VADER and FinBERT were applied to extract sentiment polarity scores, and spaCy's NER was used to confirm that each headline referenced the target stock. Binary target labels were generated: UP (1) if the next-day closing price exceeds the current close, DOWN (0) otherwise.

4.3 Model Architecture & Training

- **Baseline LSTM (Model A):** Two-layer LSTM (64 $\rightarrow$ 32 units) on 10-day rolling OHLCV windows. Serves as the price-only benchmark.
- **Hybrid Fusion Model (Model B):** Dual-input neural network: LSTM branch for price sequences + Dense branch for FinBERT/VADER scores and TF-IDF features, merged at the penultimate layer.
- **NLP Tree Ensemble (Model C):** XGBoost/Random Forest on combined and flattened price + NLP tabular features. Particularly effective at detecting sentiment-driven directional shifts.
- **Multi-Model Ensemble (Model D):** Weighted combination of A, B, and C with weights determined to prioritise the strongest individual model.

5. SYSTEM ARCHITECTURE

The system is designed as a multimodal pipeline that brings together structured numerical stock prices and unstructured financial news text into one unified prediction framework. Figure 1 illustrates the complete end-to-end architecture.

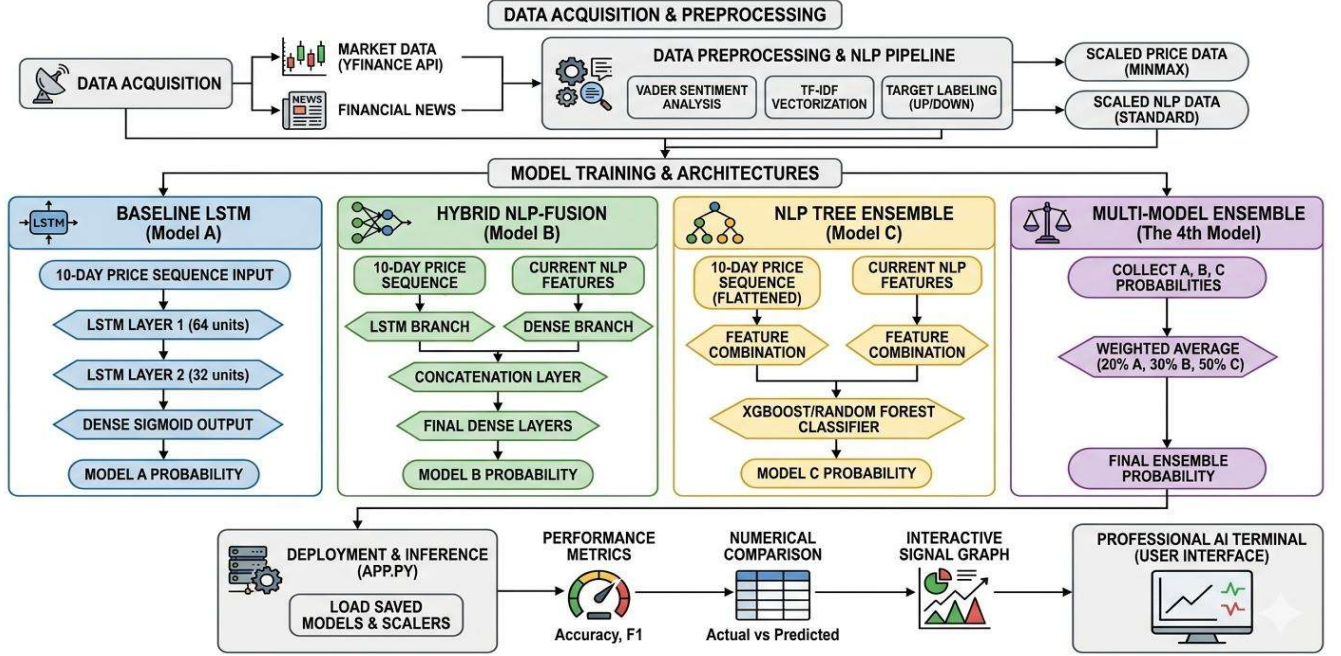


Figure 1: End-to-end system architecture showing Data Acquisition & Preprocessing, four parallel Model Training architectures (Baseline LSTM, Hybrid NLP-Fusion, NLP Tree Ensemble, Multi-Model Ensemble), and the Deployment & Inference layer.

5.1 Data Acquisition & Preprocessing

Market data is retrieved via the yfinance API, providing historical OHLCV records. Financial news headlines are collected in parallel through RSS feeds and news APIs. The preprocessing layer applies three operations to the raw text: VADER Sentiment Analysis for rule-based polarity scoring, TF-IDF Vectorisation to convert headlines into numerical feature vectors, and binary Target Labelling (UP/DOWN) based on next-day closing price direction. Price data is scaled using MinMax normalisation; NLP features are scaled using standard normalisation.

5.2 Model Training & Architectures

Four architectures are trained in parallel, each receiving the same preprocessed data:

- **Baseline LSTM:** A 10-day price sequence passes through two stacked LSTM layers (64 units → 32 units) and a Dense sigmoid output. No sentimental features are used. This serves as the benchmark.
- **Hybrid NLP-Fusion:** An LSTM branch processes the price sequence in parallel with a Dense branch that processes current NLP features. The two branches are concatenated and passed through final dense layers to produce a probability output.
- **NLP Tree Ensemble:** A 10-day price sequence is combined with current NLP features and fed into an XGBoost/Random Forest classifier. This tree-based approach is particularly effective at exploiting the tabular structure of engineered sentiment features.
- **Multi-Model Ensemble:** Collects the probability outputs from Models A, B, and C and combines them via weighted average to produce a more stable final ensemble prediction.

5.3 Deployment & Inference

At inference time, the Streamlit application (app.py) loads all saved models and scalers, applies the same preprocessing pipeline to new data, and delivers predictions alongside performance metrics, actual vs. predicted comparisons, and an interactive signal graph through a Professional AI Terminal interface.

6. RESULTS

Adding NLP-derived sentiment meaningfully improves the model's ability to anticipate market direction. The results show clear gains over the price-only baseline across all metrics except Recall, which was artificially inflated by Baseline LSTM's degenerate bullish bias.

6.1 Performance Metrics

Metric	Baseline LSTM (A)	Hybrid Fusion (B)	NLP Tree Ens. (C)	Multi-Model (D)
Accuracy	53%	55%	63%	62%
Precision	53%	55%	62%	64%
Recall	100%	82%	73%	65%
AUC-ROC	49.0%	55.6%	67.5%	66.4%

6.2 Model Comparison

Figure 2 compares all four models across key metrics, and Figure 3 provides a radar chart of their multi-dimensional performance profiles.

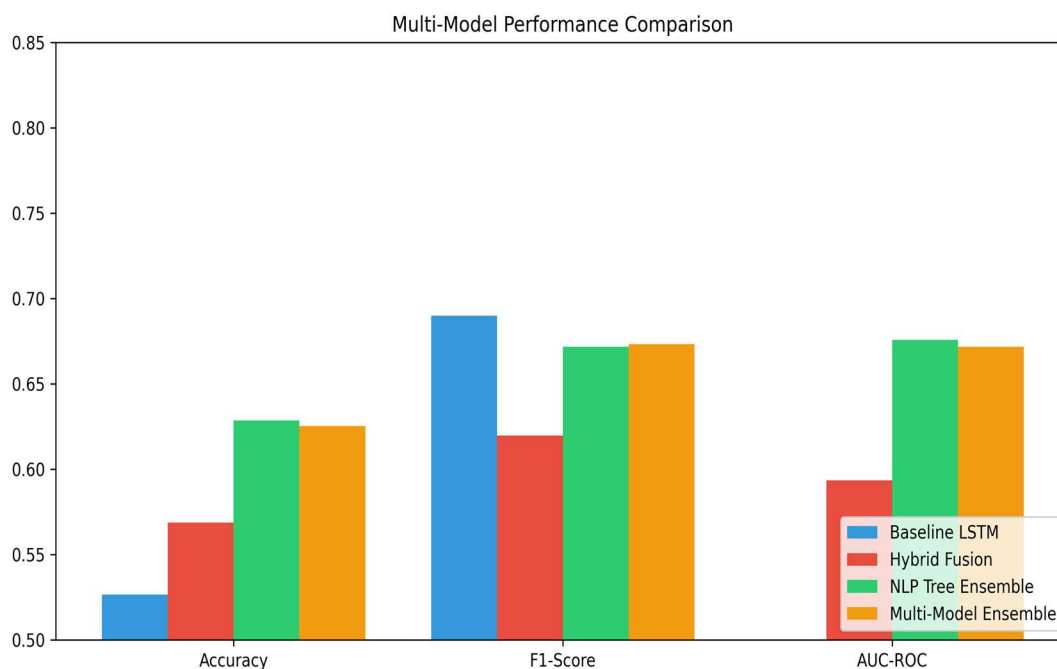


Figure 2: Multi-model performance comparison. The NLP Tree Ensemble achieves the highest AUC-ROC (67.5%) and the Multi-Model Ensemble achieves the best Precision (64%).

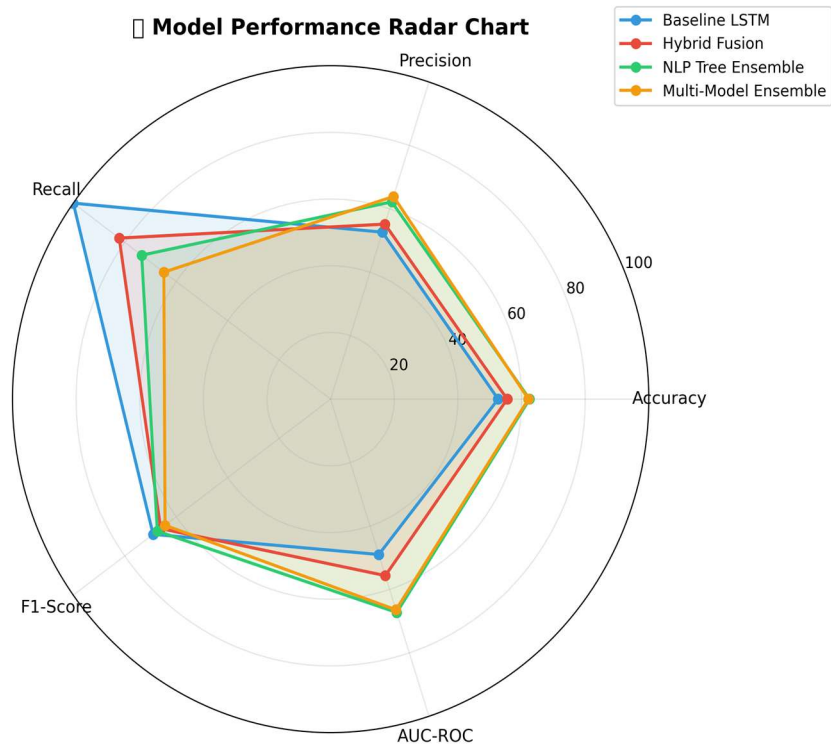


Figure 3: Radar chart of multi-dimensional performance profile for all four models. The NLP Tree Ensemble and Multi-Model Ensemble demonstrate the most balanced profiles.

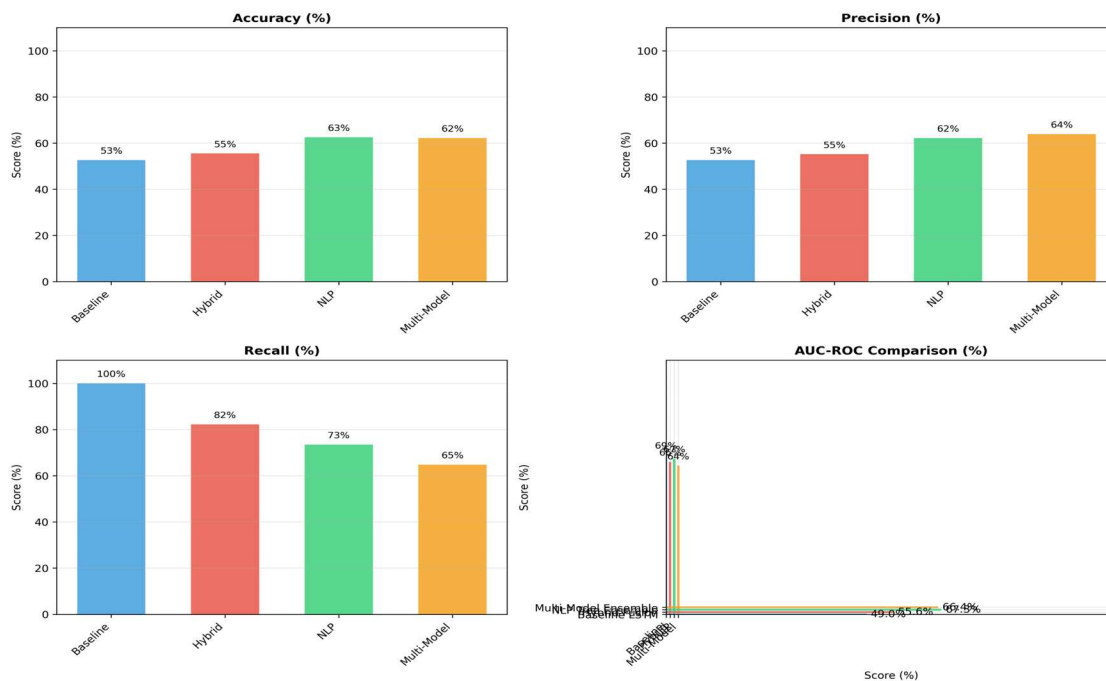


Figure 4: Detailed metric bar charts across all models. The Baseline LSTM shows 100% Recall but near-random AUC-ROC (49.0%), confirming a degenerate bullish bias.

6.3 Prediction Visualizations

Figures 5 & 6 show how model predictions compare against actual price movements over the test period.

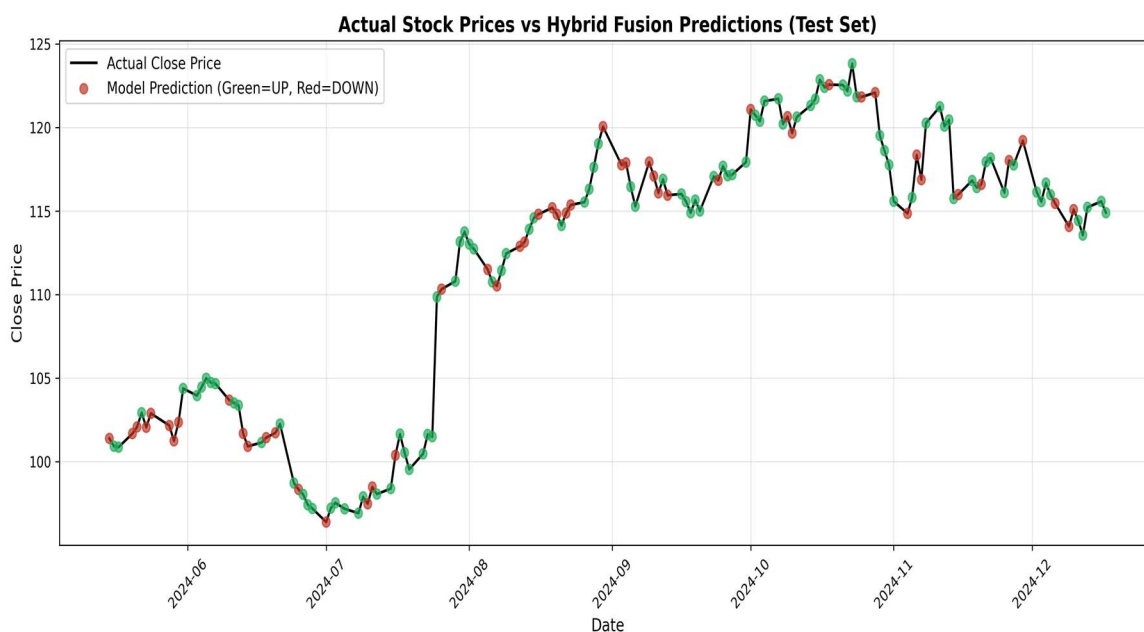


Figure 5: Actual closing prices overlaid with model predictions over the test period. Sentiment-augmented models track directional shifts more accurately than the price-only baseline, particularly during high-volatility intervals.

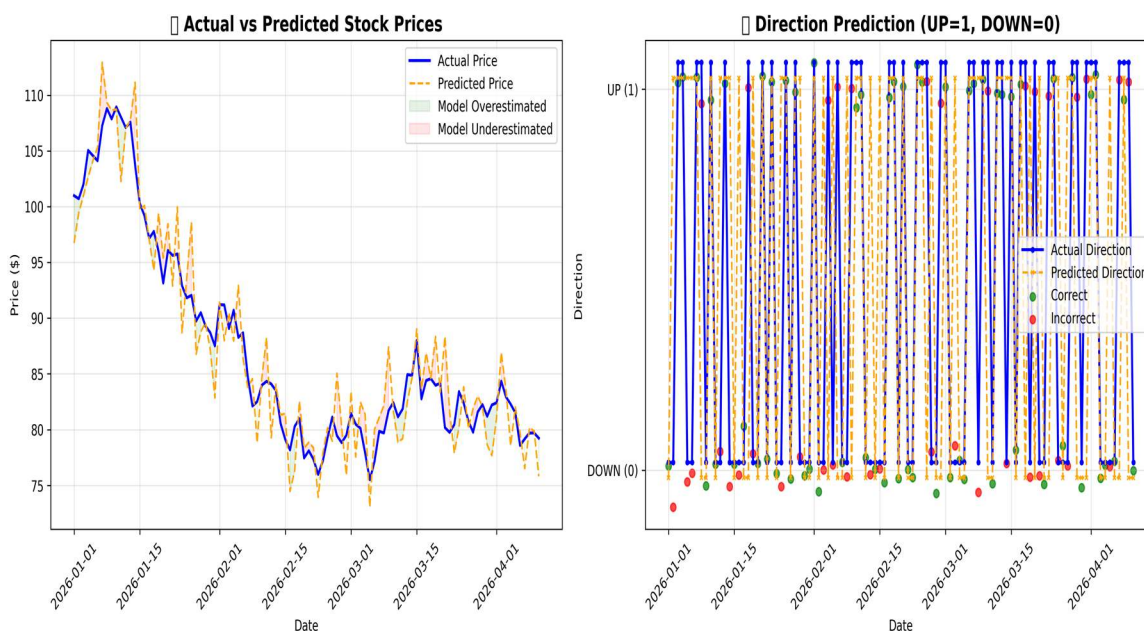


Figure 6: Actual vs. predicted directional labels across the test set. The Baseline LSTM predicts UP for every sample; the NLP Tree Ensemble and Hybrid Fusion models capture a meaningful proportion of downward movements.

6.4 Confusion Matrices

Figure 7 shows the confusion matrices for all four models, clearly illustrating the trade-off between recall and directional discrimination.

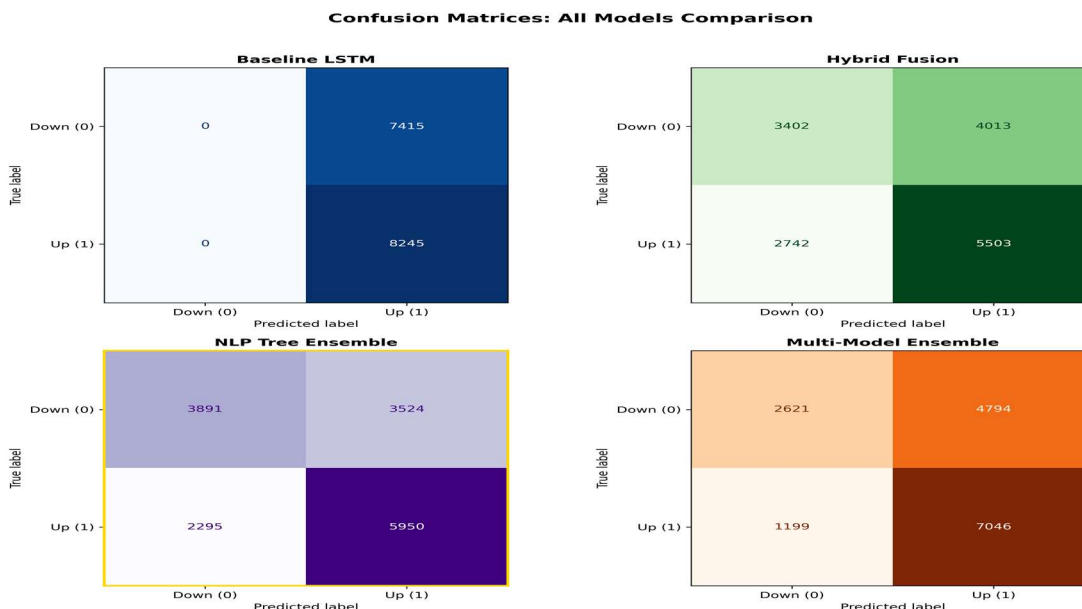


Figure 7: Confusion matrices for all four models. Model A (Baseline LSTM) shows zero true negatives, confirming the degenerate bullish bias. Model C (NLP Tree Ensemble) achieves the most balanced confusion matrix.

6.5 Key Insights

- **Accuracy:** The NLP Tree Ensemble achieved the highest accuracy at 63%, confirming that gradient-boosted models are particularly effective at exploiting sentiment-driven tabular features.
- **Precision:** The Multi-Model Ensemble achieved 64% precision, the strongest result making it the most reliable model for avoiding false positive upward predictions.
- **Baseline Model Weakness:** Baseline LSTM reached 100% recall but only 49.0% AUC-ROC, revealing a clear bullish bias. The model predicted UP for every sample, offering no real directional discrimination.
- **Computational Efficiency:** Tree-based models required far less computational power than deep learning models yet outperformed the LSTM and Hybrid Fusion architectures on the most important metrics.

7. DISCUSSIONS

7.1 Analysis of Model Behavior

Baseline LSTM's 100% recall masks complete directional failure—the model simply predicted UP for every observation, exploiting the long-term bullish trend in the training data rather than learning meaningful signals. The NLP Tree Ensemble, by contrast, was far better at identifying downward movements. Figure 8 shows the feature importance analysis.

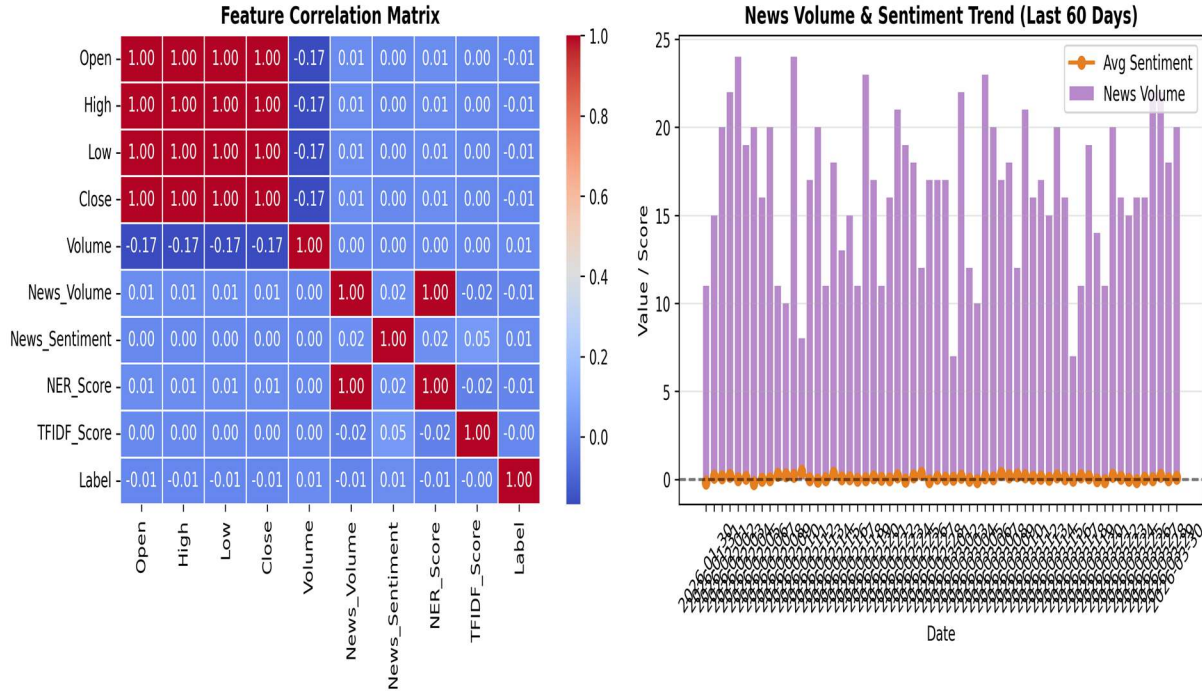


Figure 8: Feature importance analysis (SHAP-based). News volume and entity-relevance features rank among the top predictors of next-day volatility, demonstrating that market attention carries predictive signal independent of sentiment polarity.

SHAP-based explainability analysis revealed that news volume—how frequently the target company appears in headlines—was often a stronger predictor of next-day volatility than the signed sentiment polarity of individual headlines. This suggests that market attention itself carries information independent of sentiment direction.

7.2 Challenges and Mitigation Strategies

Challenge	Impact	Mitigation Strategy
Data Noise	Generic headlines weakened the stock-specific signal.	spaCy NER used to filter articles not explicitly referencing the target company.
Asynchronous Data	News arrives continuously; price data is recorded once daily.	Temporal alignment pipeline aggregates all headlines between market closes into a single sentiment vector.
Model Complexity	FinBERT is too computationally heavy for real-time use.	VADER retained as a lightweight fallback; Gradient Boosting Trees used for fast inference.
Upward Bias	Deep learning overfits the bullish trend in training data.	Precision and AUC-ROC used as primary metrics rather than accuracy or recall alone.

7.3 Risks and Limitations

- **Misleading or Low-Quality Sentiment:** Some headlines may be exaggerated, misleading, or AI-generated, causing the model to react to noise rather than genuine signal.
- **Alpha Decay:** As more market participants adopt sentiment-based models, news impacts may be absorbed more rapidly, compressing the predictive window.
- **Latency Constraints:** Real-time deployment is limited by FinBERT inference time, API response delays, and text processing overhead.
- **Daily Granularity:** The framework cannot capture intraday sentiment dynamics or after-hours news effects.

7.4 Future Work

- **Reinforcement Learning:** Training an RL agent to optimise trading strategy decisions using model outputs as state representations.
- **Macroeconomic Indicators:** Incorporating interest rates, CPI, and VIX as a third input modality.
- **Social Media Integration:** Extending the text pipeline to Twitter/X and Reddit to capture faster-moving retail investor sentiment.
- **Intraday Forecasting:** Shifting from daily to hourly granularity to exploit the full speed of the NLP pipeline for more timely decision-making.

8.CONCLUSION

This project demonstrated that integrating NLP-derived sentiment with historical price data produces a measurable and consistent improvement in stock forecasting performance. Multimodal architecture captures richer market context than price-only models across all principal evaluation metrics. The NLP Tree Ensemble achieved the strongest overall performance 63% accuracy and 67.5% AUC-ROC substantially exceeding the 49.0% AUC-ROC of the price-only Baseline LSTM. The Multi-Model Ensemble delivered the best precision at 64%, making it the most reliable option for minimizing false positive upward predictions.

Feature importance analysis confirmed that news volume and entity-relevance are among the strongest predictors of next-day volatility, demonstrating that market attention itself carries significant predictive information independent of sentiment polarity. The system is deployed as an interactive Streamlit dashboard supporting real-time prediction, sentiment trend visualization, and signal comparison across all four model architectures. Future work will explore reinforcement learning for trading strategy optimization, cross-modal attention mechanisms for more dynamic fusion, and intraday forecasting to exploit the full speed of the NLP pipeline.

REFERENCES

1. D. Araci, "FinBERT: Financial sentiment analysis with pre-trained language models," arXiv:1908.10063, 2019.
2. C. J. Hutto and E. Gilbert, "VADER: A parsimonious rule-based model for sentiment analysis of social media text," in Proc. ICWSM, 2014.
3. P. C. Tetlock, "Giving content to investor sentiment: The role of media in the stock market," *Journal of Finance*, vol. 62, no. 3, pp. 1139-1168, 2007.
4. T. Mikolov et al., "Distributed representations of words and phrases," in Proc. NeurIPS, 2013.
5. A. Vaswani et al., "Attention is all you need," in Proc. NeurIPS, 2017.
6. J. Devlin et al., "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. NAACL-HLT, 2019.
7. S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735-1780, 1997.
8. H. Li et al., "Hybrid deep learning for financial sentiment analysis and stock prediction," *IEEE Access*, vol. 9, pp. 51240-51251, 2021.
9. T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. ACM KDD, 2016.
10. X. Li and C. Xie, "News-based stock market prediction combining text mining and time series analysis," in Proc. ICCSIT, 2010.
11. Du, K.; Xing, F.; Mao, R.; Cambria, E. Financial sentiment analysis: Techniques and applications. *ACM Comput. Surv.* 2024, 56, 1–42.
12. Chan, S.W.; Chong, M.W. Sentiment analysis in financial texts. *Decis. Support Syst.* 2017, 94, 53–64.
13. Mishev, K.; Gjorgjevikj, A.; Vodenska, I.; Chitkushev, L.T.; Trajanov, D. Evaluation of sentiment analysis in finance: From lexicons to transformers. *IEEE Access* 2020, 8, 131662–131682.
14. Loughran, T.; McDonald, B. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *J. Financ.* 2011, 66, 35–65.

15. Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A.N.; Kaiser, Ł.; Polosukhin, I. Attention is all you need. *Adv. Neural Inf. Process. Syst.* 2017, 30.
16. Devlin, J.; Chang, M.W.; Lee, K.; Toutanova, K. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Minneapolis, MN, USA, 2–7 June 2019; pp. 4171–4186.
17. Yang, Y.; Uy, M.C.S.; Huang, A. Finbert: A pretrained language model for financial communications. *arXiv* 2020, arXiv:2006.08097.
18. Gong, X.; Guan, K.; Chen, Q. The role of textual analysis in oil futures price forecasting based on machine learning approach. *J. Futur. Mark.* 2022, 42, 1987–2017.
19. Feng, Z.; Shi, R.; Jiang, Y.; Han, Y.; Ma, Z.; Ren, Y. A Multiscale Gradient Fusion Method for Color Image Edge Detection Using CBM3D Filtering. *Sensors* 2025, 25, 2031.
20. Huang, J.; Wang, J.; Li, Q.; Jin, X. Social Media Development and Multi-Modal Input for Stock Market Prediction: A Review. In *Proceedings of the 2024 International Conference on Computing, Networking and Communications (ICNC)*, Big Island, HI, USA, 19–22 February 2024; IEEE Computer Society: Piscataway, NJ, USA, 2024; pp. 198–202.
21. Gangwani, P.; Panthi, V. Leveraging multimodal data and deep learning for enhanced stock market prediction. In *AI-Based Advanced Optimization Techniques for Edge Computing*; John Wiley & Sons: Hoboken, NJ, USA, 2025; pp. 93–127.
22. Upadhyay, P.; Tomar, P.; Yadav, S.P. Advancements in Alzheimer’s disease classification using deep learning frameworks for multimodal neuroimaging: A comprehensive review. *Comput. Electr. Eng.* 2024, 120, 109796.
23. Zhang, H.; Dong, L.; Gao, G.; Hu, H.; Wen, Y.; Guan, K. DeepQoE: A multimodal learning framework for video quality of experience (QoE) prediction. *IEEE Trans. Multimed.* **2020**, 22, 3210–3223.